

Zach Lawrence

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Research Interests

AI for scientific computing, surrogate modeling, time series analysis, and machine learning theory.

Education

University of Wisconsin–Madison

September 2026 – Present

Ph.D. in Computer Science, Minor in Mathematics (expected 2031)

Madison, WI. Advisor: Prof. Misha Khodak

University of California, San Diego

September 2022 – June 2026

B.S. in Computer Science, Minors in Physics and Mathematics

La Jolla, CA. GPA: 3.87/4.00. Quarterly Provost Honors (12 of 12 quarters)

Publications

1. Y. Cao, S. Lai, H. Huang, Y. Zhang, **Z. Lawrence**, R. Bhakta, I. Thomas, M. Cao, C. Tsai, Z. Zhou, Y. Zhao, H. Liu, A. Marinoni, A. Arefiev, and R. Yu. “SimulCost: A Cost-Aware Benchmark and Toolkit for Automating Physics Simulations with LLMs.” *International Conference on Machine Learning (ICML)*, 2026. arXiv:2603.20253.
2. **Z. Lawrence**, J. Yao, and C. Qin. “Deep Learning for Modeling and Dispatching Hybrid Wind Farm Power Generation.” *IEEE International Conference on Data Mining Workshops (ICDMW)*, 2025. arXiv:2512.00728.

Conference Presentations

1. **Z. Lawrence** and J. Yao. “Deep Learning for Modeling and Dispatching Hybrid Wind Farm Power Generation.” Oral presentation, IEEE International Conference on Data Mining Workshops (ICDMW), Washington, D.C., November 12, 2025.

Poster Presentations

1. Y. Cao, S. Lai, H. Huang, Y. Zhang, **Z. Lawrence**, R. Bhakta, I. Thomas, M. Cao, C. Tsai, Z. Zhou, Y. Zhao, H. Liu, A. Marinoni, A. Arefiev, and R. Yu. “SimulCost: A Cost-Aware Benchmark and Toolkit for Automating Physics Simulations with LLMs.” *International Conference on Machine Learning (ICML)*, 2026.
2. **Z. Lawrence** and J. Yao. “Deep Learning for Strategic Power Generation for Hybrid Wind Farms.” Summer Undergraduate Research Symposium, Vancouver, WA, August 1, 2025.
3. J. Yao and **Z. Lawrence**. “Atmospheric Data-Driven Time Series Modeling of Wind Power Generation.” Summer Undergraduate Research Symposium, Vancouver, WA, August 1, 2025.
4. **Z. Lawrence**, G. Wrye, A. Panthi, and R. Rai. “Mangrove Monitoring.” Engineers for Exploration REU Symposium, Escondido, CA, August 2024.

Research Experience

Khodak Group, University of Wisconsin–Madison

August 2026 – Present

Graduate Researcher, advised by Prof. Misha Khodak

Madison, WI

- Designed a novel optimization technique for PDE preconditioning that achieves up to a 2x solver convergence speed-up in initial experiments on convective PDE systems.

Rose Yu Spatiotemporal Lab, University of California, San Diego**April 2025 – Present***Undergraduate Researcher, advised by Prof. Yadi Cao and Prof. Rose Yu*

La Jolla, CA

- Built a Bayesian active learning framework for deep learning surrogate models of a high-fidelity turbulent-transport code used in nuclear fusion simulation, in collaboration with General Atomics (paper in progress).
- Designed modular acquisition algorithm based on state-of-the-art methods for multi-fidelity active learning under heavy data imbalance, which reduced training data required to reach baseline performance to 8.6% data utilization.
- Engineered a Docker and Kubernetes based job scheduler for automatically running nuclear fusion simulations on the NSF-funded Nautilus computing cluster.
- Co-authored SimulCost (ICML 2026), a cost-aware benchmark and toolkit for LLM agents that tune physics simulators: integrated a fusion simulator and a physics-informed DIII-D-based dataset, and ran parameter-scan baselines and LLM evaluations.

Washington State University Vancouver**June 2025 – August 2025***NSF REU Undergraduate Researcher, advised by Prof. Chris Qin*

Vancouver, WA

- Designed and evaluated classical probabilistic and deep learning models for estimating power generation and optimizing storage dispatch for wind farms with integrated energy storage.
- Built and trained an NQF-RNN-based probabilistic wind power generation model driven by atmospheric conditions; cut RMSE by 9.5% and improved power-curve similarity by 18.9% over PNNL benchmarks.
- Designed COVE-NN, an LSTM-based recurrent neural network dispatch policy model trained with a custom unsupervised loss to maximize energy value; reduced annual cost of valued energy (COVE) by 32.3% versus baseload dispatch in a 43-year case study.
- Full paper accepted and presented at the REU workshop of the 25th IEEE International Conference on Data Mining (ICDM 2025).

Di Ventra Group, University of California, San Diego**February 2025 – June 2025***Undergraduate Researcher, advised by Prof. Massimiliano Di Ventra and Dr. Yuanhang Zhang*

La Jolla, CA

- Developed a deep learning methodology for probabilistic forecasting of fractional Brownian motion
- Applied dynamical systems methods to analyze autocorrelation and cross-correlation of LSTM memory cells toward an explanation of stochasticity and its impact on training dynamics.

E4E Mangrove Monitoring Team, University of California, San Diego**April 2025 – June 2026***Undergraduate Researcher, advised by Prof. Ryan Kastner and Prof. Curt Schurgers*

La Jolla, CA

- Designed and trained a lightweight convolutional neural network (CNN) based on the MobileNetV2 architecture for bird-call classification.
- Profiled model performance under multiple post-training quantization schemes and deployed the model to an embedded device using code generation tools, achieving 83% precision on test data.

E4E Acoustic Species Team, University of California, San Diego**April 2023 – June 2025***Project Lead, advised by Prof. Ryan Kastner and Prof. Curt Schurgers*

La Jolla, CA

- Led a team of undergraduate developers building a web platform that maps mangroves in aerial imagery using a semantic segmentation model, in collaboration with the Scripps Institution of Oceanography.
- Implemented continuous user-facing progress feedback via WebSocket connections between an Express.js central web server, a Flask image classification server, and a React frontend.
- NSF REU Undergraduate Researcher (June 2024 – August 2024): designed, provisioned, and deployed AWS infrastructure (EC2, Fargate, ECS, S3) following least-privilege principles, and served the segmentation model from a Flask endpoint for an end-to-end web pipeline.

Honors and Awards

- NSF Research Experiences for Undergraduates (REU) award, Summer 2025.
- NSF Research Experiences for Undergraduates (REU) award, Summer 2024.
- Quarterly Provost Honors, University of California, San Diego (12 of 12 quarters), 2022 – 2026.

Technical Skills

- **Programming Languages:** Python, C, C++, Java, JavaScript, TypeScript, MATLAB, Bash, Verilog, PHP
- **Machine Learning and Scientific Computing:** PyTorch, TensorFlow, NumPy, Hydra (HDF5)
- **Systems and Infrastructure:** Docker, Kubernetes, Slurm, AWS (EC2, ECS, Fargate, S3), Linux, Git
- **Web and Software Engineering:** React, Express.js, Flask, WebSockets, REST APIs

Selected Coursework

Graduate: Machine Learning, Computational Mathematics, Quantum Computing.

Undergraduate: Deep Learning, Classical Machine Learning, Neuromorphic Computing, Computational Stochastics, Computational Physics, Mathematical Modeling, Numerical Linear Algebra, Statistical Methods, Algorithm Design, Data Structures, Computer Architecture, Compiler Construction, Software Engineering, Computer Graphics, Cryptography, Functional Programming, Embedded Systems Project, Discrete Mathematics, Linear Algebra, Calculus I–III, Vector Calculus, Differential Equations, Classical Mechanics, Fluid Mechanics, Electricity and Magnetism, Optics, Quantum Mechanics.